ExoRehab

EXOSKELETON REHABILITATION

SMA Negeri 1 Surakarta

SMA Negeri 3 Surakarta

A Sustainable Smart Exoskeleton for Post-Stroke Hemiparesis Rehabilitation

with AI-Driven Adaptive Assistance and Real-Time IoT Monitoring

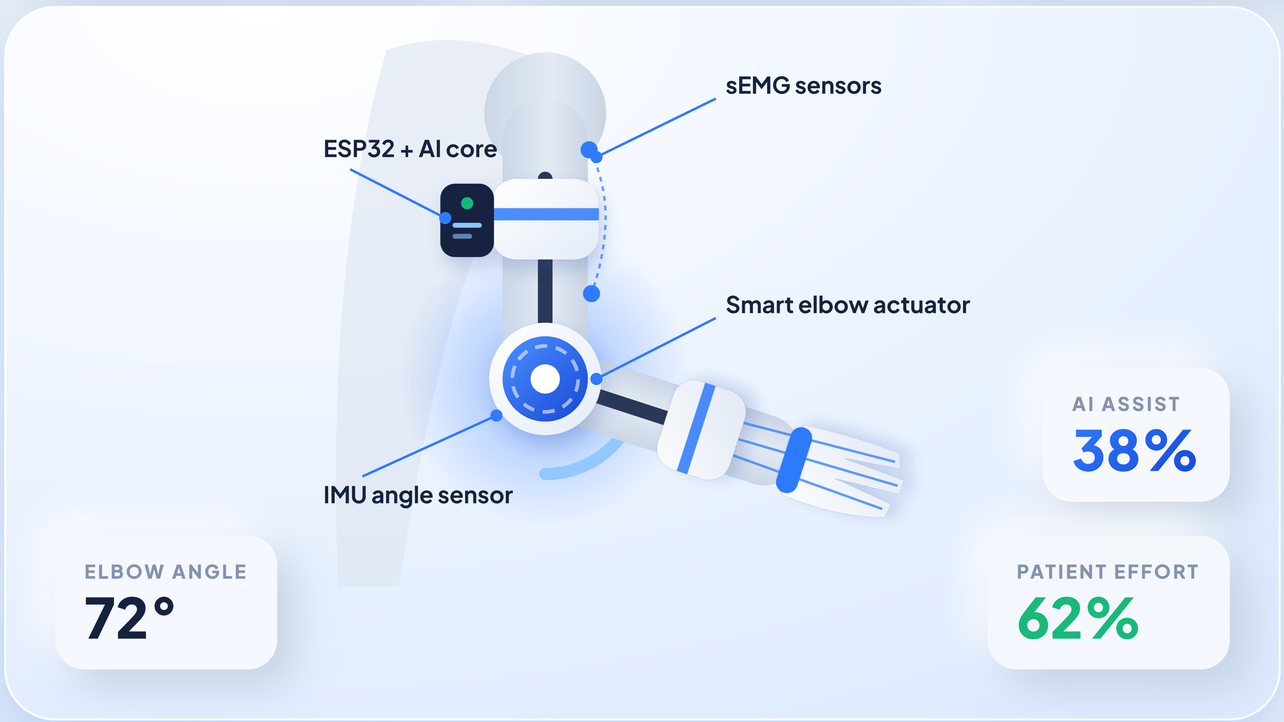
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1 BACKGROUND

Stroke takes away movement. Therapy is hard to reach.

Hemiparesis — weakness on one side of the body — is the most common motor effect of stroke. Recovery depends on **hundreds of guided repetitions**, yet therapists, clinic time and travel are limited, especially once patients return home. Most existing rehabilitation robots are large, expensive and stay in hospitals.

12.2M new strokes worldwide every year

~80% of survivors live with motor weakness on one side

#1 cause of death in Indonesia is stroke

Sources: World Stroke Organization, 2022 · Hemiparesis prevalence, stroke literature · Indonesian Ministry of Health / IHME

2 HOW IT WORKS

A closed loop that follows the patient's intention.

1 Sense
sEMG, IMU and force sensors read muscle intent, joint angle and grip in real time.

2 Think
An on-device AI model estimates patient effort and fatigue every 50 ms.

3 Assist
The actuator adds only the torque the patient is missing — assist-as-needed.

4 Monitor
ESP32 streams sessions over IoT to therapists and family dashboards.

Repeats every 50 ms ↻

3 AI ADAPTIVE ASSISTANCE

Help only as much as needed.

$$\tau_{motor} = (1 - \hat{e}) \cdot \tau_{task}$$

ê = patient effort estimated by AI from sEMG + joint kinematics

- A lightweight model on the ESP32 classifies muscle activity and movement into an **effort score**.
- As the patient improves, assistance **fades automatically** — keeping therapy challenging but safe.
- Fatigue or spasm detection **raises assistance or soft-stops** the actuator.

4 REAL-TIME IOT MONITORING APP

Every session, live and measurable.



Live digital twin · Progress analytics · Care-team monitor

5 SYSTEM ARCHITECTURE

From muscle signal to care team.

Sensors
sEMG · IMU · force · heart rate

▼

ESP32-S3 edge AI
effort estimation · control · safety

▼

Elbow actuator
assist-as-needed torque

IoT cloud
MQTT · TLS

▼

Therapist
live dashboard

Patient & family
progress · alerts

6 SUSTAINABILITY

Built to last, built to be fixed.

Recycled materials
3D-printed recycled PETG frame, washable replaceable cuffs.

Modular & repairable
Cuffs, battery and actuator snap apart for repair.

Low-power
Edge AI on a microcontroller, deep sleep, USB-C charging.

Fewer trips
Home therapy with remote monitoring reduces travel.

7 IMPACT & RESEARCH PLAN

Toward the SDGs.

- **3** Good Health & Well-being
- **9** Industry, Innovation & Infrastructure
- **10** Reduced Inequalities
- **12** Responsible Consumption

01 Literature & patient criteria

02 Design & prototype

03 AI model & IoT app

04 Testing & evaluation

Student research prototype — not a certified medical device. App data shown is simulated.